1 Two identical light sources are viewed from a distance, as shown in Fig 1.1. When the angle θ between the light sources is large, they are seen as separate.

For Examiner's Use

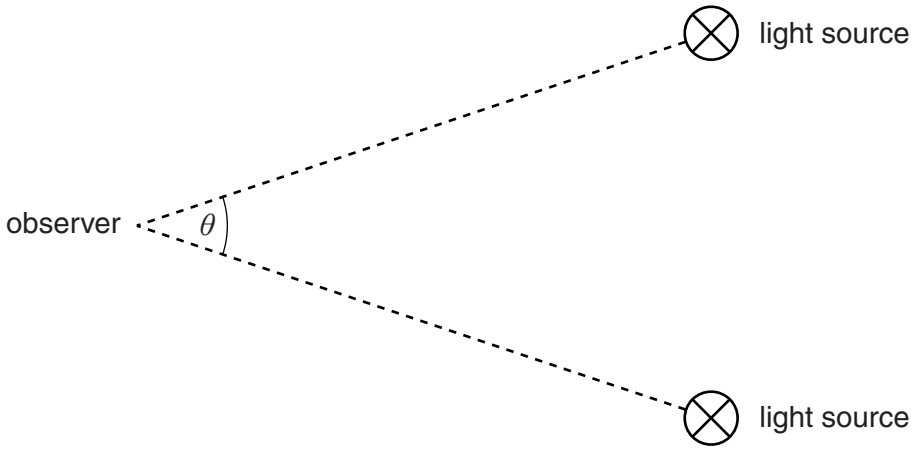

Fig 1.1 (not to scale)

The sources are moved closer together. At a particular angle θ_1 the two sources appear as a single source.

It is suggested that θ_1 is directly proportional to the wavelength λ of the light from the sources.

Design a laboratory experiment using two light sources to test the relationship between θ_1 and λ. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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